Bureau
International des
Poids et
Mesures

# Le système international d'unités
# The International System of Units

**9 édition 2003**

Annexe 2
**Appendix 2**

## Valeurs recommandées des fréquences étalons destinées à la mise en pratique de la définition du mètre et aux représentations secondaires de la seconde

## Recommended values of standard frequencies for applications including the practical realization of the metre and secondary representations of the second

Partie 2.14
**Part 2.14**

## Iodine ($\lambda \approx 612\,\mathrm{nm}$)

## Iodine ($\lambda \approx 612\,\mathrm{nm}$)

**Bureau International**
**des Poids et Mesures**

# Recommended values of standard frequencies for applications including the practical realization of the metre and secondary representations of the second
# Part 2.14: Iodine ($\lambda \approx 612\ \mathrm{nm}$)

**9th edition 2003**

Appendix 2

## Absorbing molecule $^{127}I_2$, $a_7$ component, R(47) 9-2 transition[1]

[1] All transitions in $I_2$ refer to the $B^3\Pi\ 0_u^+ - X^1\ \Sigma_g^+$ system.

# 1. CIPM recommended values

The values $f = 489\,880\,354.9$ MHz
$\lambda = 611\,970\,770.0$ fm

with a relative standard uncertainty of $3 \times 10^{-10}$ apply to the radiation of a He-Ne laser stabilized with an iodine cell, within or external to the laser, having a cold-finger temperature of $(-5 \pm 2)$ °C[1].

[1] For the specification of operating conditions, such as temperature, modulation width and laser power, the symbols ± refer to a tolerance, not an uncertainty.

# 2. Source data

Adopted value

$f = 489\,880\,354.93(15)$ MHz

$u_c/y = 3.0 \times 10^{-10}$

for which:

$\lambda = 611\,970\,769.97(18)$ fm

$u_c/y = 3.0 \times 10^{-10}$

calculated from

| $f$/kHz | $u_c/y$ | source data |
|---|---|---|
| 489 880 354 979 | $1 \times 10^{-10}$ | Section 2.1 |
| 489 880 354 728 | $2.1 \times 10^{-10}$ | Section 2.2 |
| 489 880 355 026 | $8.3 \times 10^{-11}$ | Section 2.3 |
| 489 880 355 062 | $3.0 \times 10^{-10}$ | Section 2.4 |
| 489 880 358 850 | $8.5 \times 10^{-11}$ | Section 2.5 |
| Unweighted mean: | $f = 489\,880\,354\,929$ kHz | |

Other available values having relative uncertainties higher than $3.0 \times 10^{-10}$ have not been used. The relative standard uncertainty calculated from the dispersion of the six values is $2.8 \times 10^{-10}$, which the CCL preferred to round up to $3.0 \times 10^{-10}$.

Source data

## 2.1.

Reference [1] gives

$f_{a7}/f_i = 1.034\,349\,072\,43$

$u_c/y = 1 \times 10^{-10}$.

Using the recommended value of the absorbing molecule $^{127}I_2$, $a_{16}$ or f component, R(127) 11-5 transition (see iodine at $\lambda \approx 633$ nm and frequency differences listed in corresponding Table 1, see p. 9) one obtains

$f_i = 473\,612\,214\,712$ kHz

$u_c/y = 2.2 \times 10^{-11}$,

one calculates

$f_{a7} = 489\,880\,354\,979$ kHz

$u_c/y = 1 \times 10^{-10}$.

## 2.2.

Reference [2] gives

$f_{a7}/f_i = 1.034\,349\,071\,90$

$u_c/y = 2.1 \times 10^{-10}$.

Using the recommended value of the absorbing molecule $^{127}I_2$, $a_{16}$ or $f$ component, R(127) 11-5 transition (see iodine at $\lambda \approx 633$ nm and frequency differences listed in corresponding Table 1, see p. 9) one obtains

$f_i = 473\,612\,214\,712$ kHz
$u_c/y = 2.2 \times 10^{-11}$,

one calculates

$f_{a7} = 489\,880\,354\,728$ kHz
$u_c/y = 2.1 \times 10^{-10}$.

## 2.3.

Bönsch et al. [3] give $\lambda_{b15}/\lambda_i = 0.966\,791\,921\,43$
$u_c/y = 8 \times 10^{-11}$.

Using the recommended value of the absorbing molecule $^{127}I_2$, $a_{16}$ or $f$ component, R(127) 11-5 transition (see iodine at $\lambda \approx 633$ nm and frequency differences listed in corresponding Table 1, see p. 9) one obtains

$f_i = 473\,612\,214\,712$ kHz
$u_c/y = 2.2 \times 10^{-11}$,

one calculates

$f_{b15} = 489\,880\,194\,708$ kHz
$u_c/y = 8.3 \times 10^{-11}$.

From the measured value (see Table 4 below)
$f_{b15} - f_{a7} = -160\,318$ kHz
$u_c = 3$ kHz

one calculates

$f_{a7} = 489\,880\,355\,026$ kHz
$u_c/y = 8.3 \times 10^{-11}$.

## 2.4.

Vitushkin et al. [4] give $\lambda_d/\lambda_{a7} = 1.034\,348\,712$
$u_c/y = 3 \times 10^{-10}$.

Using the recommended value of the absorbing molecule $^{127}I_2$, $a_{16}$ or f component, R(127) 11-5 transition (see iodine at $\lambda \approx 633$ nm and frequency differences listed in corresponding Table 1, see p. 9) one obtains

$f_d = 473\,612\,379\,828$ kHz
$u_c/y = 2.2 \times 10^{-11}$,

one calculates

$f_{a7} = 489\,880\,355\,062$ kHz

$u_c/y = 3.0 \times 10^{-10}$.

## 2.5.

Himbert et al. [5] give $f_{a13} = 489\,880\,604\,541$ kHz
$u_c = 88$ kHz.

This value is a result of the frequency ratio $f_{a13}/f_e$, to which the recommended value adopted by the CIPM in 1983 [6], [7] was applied, i.e. $f_i = 473\,612\,214.8$ MHz. (see iodine at $\lambda \approx 633$ nm and frequency differences listed in corresponding Table 1, see p. 9)

$f_e - f_i = 152\,255$ kHz
$u_c = 5$ kHz,

one obtains

$f_e = 473\,612\,367\,055$ kHz,

and hence

$f_{a13}/f_e = 1.034\,349\,267$
$u_c/y = 8 \times 10^{-11}$.

Using the recommended value of the absorbing molecule $^{127}I_2$, $a_{16}$ or f component, R(127) 11-5 transition (see iodine at $\lambda \approx 633$ nm and frequency differences listed in corresponding Table 1, see p. 9) one obtains

$f_e = 473\,612\,366\,967$ kHz
$u_c/y = 2.2 \times 10^{-11}$,

one calculates

$f_{a13} = 489\,880\,604\,450$
$u_c/y = 8.3 \times 10^{-11}$.

Knowing the frequency difference (see Table 1, see p. 9)
$f_{a7} - f_{a13} = -249\,600$ kHz
$u_c = 10$ kHz,

one obtains

$f_{a7} = 489\,880\,354\,850$
$u_c/y = 8.5 \times 10^{-11}$.

# 3. Absolute frequency of the other transitions related to those adopted as recommended and frequency intervals between transitions and hyperfine components

These tables replace those published in BIPM Com. Cons. Long., 2001, **10**, 184-187 and *Metrologia*, 2003, **40**, 127-128.

The notation for the transitions and the components is that used in the source references. The values adopted for the frequency intervals are the weighted means of the values given in the references.

For the uncertainties, account has been taken of:

- the uncertainties given by the authors;
- the spread in the different determinations of a single component;
- the effect of any perturbing components;
- the difference between the calculated and the measured values.

In the tables, $u_c$ represents the estimated combined standard uncertainty ($1\sigma$).

All transitions in molecular iodine refer to the B-X system.

**Table 1**

$\lambda \approx 612$ nm $^{127}I_2$ R(47) 9-2

| $a_n$ | $x$ | $[f(a_n)-f(a_7)]$/MHz | $u_c$/MHz | $a_n$ | $x$ | $[f(a_n)-f(a_7)]$/MHz | $u_c$/MHz |
|---|---|---|---|---|---|---|---|
| $a_1$ | u | −357.16 | 0.02 | $a_{12}$ | j | 219.602 | 0.006 |
| $a_2$ | t | −333.97 | 0.01 | $a_{13}$ | i | 249.60 | 0.01 |
| $a_3$ | s | −312.46 | 0.02 | $a_{14}$ | h | 284.30 | 0.01 |
| $a_4$ | r | −86.168 | 0.007 | $a_{15}$ | g | 358.37 | 0.03 |
| $a_5$ | q | −47.274 | 0.004 | $a_{16}$ | f | 384.66 | 0.01 |
| $a_6$ | p | −36.773 | 0.003 | $a_{17}$ | e | 403.76 | 0.02 |
| $a_7$ | o | 0 | — | $a_{18}$ | d | 429.99 | 0.02 |
| $a_8$ | n | 81.452 | 0.003 | $a_{19}$ | c | 527.16 | 0.02 |
| $a_9$ | m | 99.103 | 0.003 | $a_{20}$ | b | 539.22 | 0.02 |
| $a_{10}$ | l | 107.463 | 0.005 | $a_{21}$ | a | 555.09 | 0.02 |
| $a_{11}$ | k | 119.045 | 0.006 | | | | |

Frequency referenced to $a_7$, R(47) 9-2, $^{127}I_2$: $f = 489\,880\,354.9$ MHz [8]

Ref. [9], [10], [11], [12], [13], [14]

**Table 2**

$\lambda \approx 612$ nm $^{127}I_2$ P(48) 11-3

| $b_n$ | $[f(b_n)-f(a_7)]$/MHz | $u_c$/MHz | $b_n$ | $[f(b_n)-f(a_7)]$/MHz | $u_c$/MHz |
|---|---|---|---|---|---|
| $b_1$ | −1 034.75 | 0.07 | $b_9$ | −579.91 | 0.01 |

| | | | | | |
|---|---|---|---|---|---|
| $b_2$ | −755.86 | 0.05 | $b_{10}$ | −452.163 | 0.005 |
| $b_3$ | −748.28 | 0.03 | $b_{11}$ | −316.6 | 0.4 |
| $b_4$ | −738.35 | 0.04 | $b_{12}$ | −315.8 | 0.4 |
| $b_5$ | −731.396 | 0.006 | $b_{13}$ | −297.42 | 0.03 |
| $b_6$ | −616.01 | 0.03 | $b_{14}$ | −294.72 | 0.03 |
| $b_7$ | −602.42 | 0.03 | $b_{15}$ | −160.318 | 0.003 |
| $b_8$ | −593.98 | 0.01 | | | |

Frequency referenced to $a_7$, R(47) 9-2, $^{127}I_2$: $f = 489\,880\,354.9$ MHz [8]

Ref. [9], [10], [12], [13], [14], [15]

**Table 3**

$\lambda \approx 612$ nm $^{127}I_2$ R(48) 15-5

| $c_n$ | $[f(c_n) - f(a_7)]$/MHz | $u_c$/MHz | $c_n$ | $[f(c_n) - f(a_7)]$/MHz | $u_c$/MHz |
|---|---|---|---|---|---|
| $c_1$ | −513.83 | 0.03 | $c_5$ | −209.96 | 0.03 |
| $c_2$ | −237.40 | 0.03 | $c_6$ | −97.74 | 0.03 |
| $c_3$ | −228.08 | 0.03 | $c_8$ | −73.92 | 0.03 |
| $c_4$ | −218.78 | 0.03 | $c_9$ | −59.30 | 0.03 |

Frequency referenced to $a_7$, R(47) 9-2, $^{127}I_2$: $f = 489\,880\,354.9$ MHz [8]

Ref. [10]

**Table 4**

$\lambda \approx 612$ nm $^{129}I_2$ P(110) 10-2

| $a_n$ | $x$ | $[f(a_n) - f(a7\{^{127}I_2\})]$ / MHz | $u_c$/MHz | $a_n$ | $x$ | $[f(a_n) - f(a_7\{^{127}I_2\})]$ / MHz | $u_c$/MHz |
|---|---|---|---|---|---|---|---|
| $a_1$ | b’ | −376.29 | 0.05 | $a_{15}$ | n | 1.61 | 0.20 |
| $a_2$ | a’ | −244.76 | 0.10 | $a_{16}$ | m | 10.63 | 0.15 |
| $a_3$ | z | −230.79 | 0.20 | $a_{17}$ | l | 15.82 | 0.20 |
| $a_4$ | y | −229.40 | 0.20 | $a_{18}$ | k | 25.32 | 0.10 |
| $a_5$ | x | −216.10 | 0.05 | $a_{19}$ | j | 49.44 | 0.15 |
| $a_6$ | w | −149.37 | 0.10 | $a_{20}$ | i | 54.66 | 0.20 |
| $a_7$ | v | −134.68 | 0.10 | $a_{21}$ | h | 69.02 | 0.10 |
| $a_8$ | u | −130.98 | 0.10 | $a_{22}$ | g | 74.47 | 0.15 |
| $a_9$ | t | −116.67 | 0.05 | $a_{23}$ | f | 110.60 | 0.10 |
| $a_{10}$ | s | −96.26 | 0.20 | $a_{24}$ | e | 153.09 | 0.20 |
| $a_{11}$ | r | −90.70 | 0.20 | $a_{25}$ | d | 154.70 | 0.20 |
| $a_{12}$ | q | −84.12 | 0.20 | $a_{26}$ | c | 163.98 | 0.20 |
| $a_{13}$ | p | −77.79 | 0.20 | $a_{27}$ | b | 166.22 | 0.20 |
| $a_{14}$ | o | −72.70 | 0.20 | $a_{28}$ | a | 208.29 | 0.10 |

Frequency referenced to $a_7$, R(47) 9-2, $^{127}I_2$: $f = 489\,880\,354.9$ MHz [8]

Ref. [16], [17], [18]

**Table 5**

| $\lambda \approx 612$ nm $^{129}I_2$ R(113) 14-4 | | | | | | | |
|---|---|---|---|---|---|---|---|
| $b_n$ | $x$ | $[f(b_n) - f(a7\{^{127}I_2\})]$ / MHz | $u_c$/MHz | $b_n$ | $x$ | $[f(b_n) - f(a_7\{^{127}I_2\})]$ / MHz | $u_c$/MHz |
| $b_{19}$ | r | −410.4 | 0.3 | $b_{28}$ | i | −289.4 | 0.5 |
| $b_{20}$ | q | −390.0 | 0.3 | $b_{29}$ | h | −273.1 | 0.3 |
| $b_{21}$ | p | −383.9 | 0.5 | $b_{30}$ | g | −255.7 | 0.5 |
| $b_{22}$ | o | −362.8 | 0.3 | $b_{31}$ | f | −247 | 5 |
| $b_{23}$ | n | −352.9 | 0.3 | $b_{32}$ | e | −237 | 5 |
| $b_{24}$ | m | −346.4 | 0.3 | $b_{33}$ | d | −223 | 5 |
| $b_{25}$ | l | −330.0 | 0.3 | $b_{34}$ | c | −198.6 | 0.3 |
| $b_{26}$ | k | −324.9 | 0.3 | $b_{35}$ | b | −193.1 | 0.3 |
| $b_{27}$ | j | −304.7 | 0.3 | $b_{36}$ | a | −187.0 | 0.3 |
| Frequency referenced to | | $a_7$, R(47) 9-2, $^{127}I_2$: $f = 489\,880\,354.9$ MHz [8] | | | | | |

Ref. [17], [18]

# References


[1] CCDM/82-34, NPL, Laser wavelength Measurements, May 1982.

[2] CCDM/82-19a, BIPM, Réponse au questionnaire CCDM/82-3.

[3] Bönsch G., Gläser M., Spieweck F., Bestimmung der Wellenlängenverhältnisse von drei $^{127}I_2$-stabilisierten Lasern bei 515 nm, 612 nm und 633 nm, *PTB Jahresbericht*, 1986, 161.

[4] Vitushkin L. F., Zakharenko Yu. G., Yvanov I. V., Leibengardt G. I., Shur V. L., Measurements of Wavelength of High-Stabilized He-Ne/$I_2$ Laser at 612 nm, *Opt. Spectr.*, 1990, **68**, 705-707.

[5] Himbert M., Bouchareine P., Hachour A., Juncar P., Millerioux Y., Razet A., Measurements of Optical Wavelength Ratios Using a Compensated Field Sigmameter, *IEEE Trans. Instrum. Meas.*, 1991, **40**, 200-203.

[6] *BIPM, Proc. Verb. Com. Int. Poids et Mesures*, 1983, **51**.

[7] Documents Concerning the New Definition of the Metre, *Metrologia*, 1984, **19**, 163-178.

[8] Recommendation CCL3 (*BIPM Com. Cons. Long.*, 10th Meeting, 2001) adopted by the Comité International des Poids et Mesures at its 91th Meeting as Recommendation 1 (CI-2002).

[9] Gläser M., Hyperfine Components of Iodine for Optical Frequency Standards *PTB-Bericht*, 1987, **PTB-Opt-25**.

[10] Razet A., Millerioux Y., Juncar P., Hyperfine Structure of the 47R(9-2), 48P(11-3) and 48R(15-5) Lines of $^{127}I_2$ at 612 nm as Secondary Standards of Optical Frequency, *Metrologia*, 1991, **28**, 309-316.

[11] Cérez P., Bennett S. J., Helium-neon laser stabilized by saturated absorption in iodine at 612 nm, *Appl. Opt.*,1979, **18**, 1079-1083.

[12] Gläser M., Properties of a He-Ne Laser at $\lambda \approx 612$ nm, Stabilized by Means of an External Iodine Absorption Cell, *IEEE Trans. Instrum. Meas.*, 1987, **IM-36**, 604-608.

[13] Bertinetto F., Cordiale P., Fontana S., Picotto G. B., Recent Progresses in He-Ne Lasers Stabilized to $^{127}I_2$, *IEEE Trans. Instrum. Meas.*, 1985, **IM-34**, 256-261.

[14] Robertsson L., Iodine-stabilized He-Ne lasers at $\lambda = 612$ nm using internal and external cells, *BIPM Proc.-Verb. Com. Int. Poids et Mesures*, 1992, **60**, 160-162.

[15] Bertinetto F., Cordiale P., Picotto G. B., Chartier J.-M., Felder R., Gläser M., Comparison Between the $^{127}I_2$ Stabilized He-Ne Lasers at 633 nm and at 612 nm of the BIPM and the IMGC, *IEEE Trans. Instrum. Meas.*, 1983, **IM-32**, 72-76.

[16] Kegung D., Gläser M., Helmcke J., $I_2$ Stabilized He-Ne Lasers at 612 nm, *IEEE Trans. Instrum. Meas.*, 1980, **IM-29**, 354-357.

[17] Ciddor P. E., Brown N., Hyperfine Spectra in Iodine-129 at 612 nm, *Opt. Commun.*, 1980, **34**, 53-56.

[18] Gläser M., Kegung D., Foth H. J., Hyperfine Structure and Fluorescence Analysis of Enriched $^{129}I_2$ at the 612 nm Wavelength of the He-Ne Laser, *Opt. Commun.*, 1981, **38**, 119-123.